

Jonathan Fung

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Experience

Sesame AI – Software Engineer, Agent Platform, Core Infra

2025 - present

- Owned the **real-time voice agent** infrastructure at Sesame, building the WebSocket and WebRTC layer that powers live AI voice conversations, enabling low-latency AI agent capabilities including streaming text and audio generation, tool calling, and turn taking.
- Managed our ML training and inference GPU clusters. Models served include custom-built Titan TTS models, turn-taking Whisper models, LLM and VAD models all self hosted on Kubernetes. Total cost of all H100 GPUs maintained totaled over **\$10M per year**.
- Developed a **self-serve ML inference platform**, enabling engineers to deploy models to Kubernetes GPU pods via a single YAML configuration. Scaled from **0 to 65 active services** within 1 year. Reduced the time to prototype ML models from **weeks to hours**.
- Migrated voice agent from a GPU VM-bound monolithic architecture to a distributed microservice model where independently scaled Kubernetes pods performed model inference, resulting in efficient GPU resource scheduling and **decreasing deploy times by 3 hours**.
- Hardened platform reliability through performance optimization, distributed tracing, testing frameworks, and adding fault tolerance to streaming systems. Accelerated eng velocity by streamlining developer workflows and formalizing oncall alerts and processes.

dYdX Trading – Software Engineer, Backend Protocol Team

2023 - 2025

- Responsible for the core e2e trading flow of dYdX protocol, a decentralized orderbook exchange. Designed and implemented the matching engine, central limit orderbook, exchange consensus mechanisms, advanced order types, and liquidations/deleveraging.
- Every trade on dYdX is executed through the matching engine, which has processed **over \$210 billion in trading volume**, yielding up to **\$2.5 million in weekly trading fees** during peak periods. Record daily volume was **\$2.1 billion**.
- Led development of a **thread-safe, real-time gRPC/websocket streaming framework** to stream orderbook events to market makers. Project doubled liquidity on the exchange.
- Responsible for long tail performance and latency work. Project enabled matching engine to process **5k orders/second** across 150 markets by supporting **5x higher throughput**.
- Managed node deployments via terraform and kubernetes. Developed first on-call rotation by adding first metrics library, monitoring, and alerting to our systems.

Scale AI – Software Engineer, Infrastructure

2020 - 2023

- Led the design and development of the task dispatch framework v2. Introduced retroactive searching, mandatory tracing, caching, and logging for increased explainability into the task dispatch service. Decreased P95 latency of dispatch endpoint by **>95%**. Project enabled a **7-digit revenue stream** opportunity by supporting a 20x higher task labelling throughput and **eliminating lock contention**.
- Designed and built Scale's **blob storage proxy microservice** to separate logical and physical data storage. Provides **per-customer encryption key** management and data lake style metadata search built on DynamoDB/Elasticsearch. Project enabled multiple significant revenue streams due to increased security from per-customer encryption keys.
- Unlocked the use of **ML linter jobs** by developing a **data linting framework** used by all data labeling queues at Scale. Linters jobs could be run either async or sync.
- Wrote coding + debugging engineering interviews, helped run university recruiting.

Pinterest – Software Engineer Intern, Observability

2019 - 2019

- Reduced infra costs by **\$1.2 Million per year** by building a by-host horizontal aggregation metrics pipeline via t-digests.
- Achieved **99% reduction** in metrics storage. Processing **8 million metrics per second**.

Stripe – Software Engineer Intern, Developer Productivity

2018 - 2018

- Implemented enums in Stripe's static ruby typechecker. Code refactored ~2000 modules to a new enum format.
- Added language server protocol support such as jump-to-def, type-on-hover to all VScode IDE and all common code editors.

Education

University of California, Berkeley

B.S. Electrical Engineering and Computer Sciences

GPA: 3.9/4.0

- Magna Cum Laude (High Honors)
- HKN EECS Honor Society
- Accel Scholar, Kleiner Perkins Engineering Fellow

Skills

Concepts

- ML Systems, Model Serving + Inference, Audio-based ML systems.
- Performance optimization, Metrics, Developer Productivity

Languages

- Golang, Python, Typescript, Ruby, Scala, Java, Bash

Technologies

- Docker, Terraform, AWS, GCP, Kubernetes, gRPC, Protobuf/Thrift
- MongoDB, Redis, Kafka, Elasticsearch, DynamoDB, MySQL